

Project: YouTube Comment Intelligence Analyzer

Role

You are a senior data scientist, NLP engineer, social media analyst, and Python developer.

Your task is to build a complete YouTube Comment Intelligence Analyzer that processes a database of YouTube comments and generates actionable audience insights.

Data Source

The comments are stored in a database.

Before writing code:

1. Inspect the database schema.
2. Identify available tables and columns.
3. Determine:
4. Comment text field
5. Video ID
6. Video title
7. Author
8. Timestamp
9. Likes
10. Reply count
11. Any additional metadata

Automatically adapt code to the discovered schema.

Objective

Analyze comments and generate:

1. Executive Summary

Provide:

- Total comments analyzed
- Time period covered
- Overall audience mood
- Most discussed topics
- Key audience concerns
- Key opportunities

2. Sentiment Analysis

Classify comments into:

- Positive
- Neutral
- Negative
- Mixed

Calculate:

- Count
- Percentage

Provide representative examples.

3. Theme Detection

Automatically cluster comments into themes.

Possible themes include:

- Praise
- Criticism
- Questions
- Suggestions
- Complaints
- Requests
- Success stories
- Technical issues
- Humor
- Community discussions

For each theme:

- Frequency
- Percentage
- Example comments
- Key insight

Use NLP-based topic modeling or clustering.

4. Topic Analysis

Identify:

- Most discussed topics
- Emerging topics

- Fast-growing discussion areas

Provide:

- Topic name
- Comment volume
- Sentiment
- Example comments

Rank topics by frequency.

5. Audience Questions

Identify comments that contain questions.

Generate:

- Frequently asked questions
- Unanswered questions
- High-interest questions

Group similar questions together.

6. Content Opportunity Detection

Identify:

- Requested videos
- Missing content
- Follow-up content opportunities
- Frequently requested explanations

Rank opportunities by demand.

7. Engagement Analysis

Determine:

- Most liked comments
- Most replied-to comments
- Most engaging discussion threads

Explain why these comments generated engagement.

8. Audience Psychology Analysis

Infer:

- Audience motivations
- Audience interests
- Audience frustrations
- Audience expectations

Support conclusions with evidence.

9. Reputation Risk Detection

Identify:

- Repeated complaints
- Negative sentiment spikes
- Trust concerns
- Controversial topics
- Potential reputation issues

Classify severity:

- Low
 - Medium
 - High
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10. Trend Analysis

Compare:

- Current period vs previous period

Detect:

- Sentiment changes
 - Topic changes
 - New audience concerns
 - New content interests
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11. Channel Growth Recommendations

Generate:

Short-Term Actions

- Comment responses
- FAQ content

- Community actions

Medium-Term Actions

- New content ideas
- Series opportunities

Long-Term Actions

- Audience strategy
 - Community-building initiatives
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Technical Requirements

Use Python.

Preferred libraries:

- pandas
- numpy
- scikit-learn
- nltk
- spacy
- sentence-transformers
- BERTopic
- matplotlib
- plotly

Create modular code:

- data_loader.py
 - preprocessing.py
 - sentiment_analysis.py
 - topic_modeling.py
 - engagement_analysis.py
 - reporting.py
 - main.py
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Output

Generate:

Interactive Dashboard

Include:

- Sentiment distribution
- Topic distribution

- Trend charts
- Audience questions
- Opportunities
- Risks

Executive Report

Export:

- HTML
- PDF
- Excel summary

Final Report Structure

1. Executive Summary
 2. Sentiment Analysis
 3. Theme Breakdown
 4. Topic Analysis
 5. Audience Questions
 6. Content Opportunities
 7. Engagement Analysis
 8. Reputation Risks
 9. Growth Recommendations
 10. Top 5 Actionable Insights
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Coding Process

Before generating final code:

1. Inspect database structure.
2. Explain discovered schema.
3. Explain proposed architecture.
4. Explain chosen NLP techniques.
5. Then generate production-ready code.

The solution should work on datasets ranging from 1,000 comments to 1,000,000+ comments.